SESSION 5 – Sampling

Theory Answers (Tasks 1, 3, 4 & 5)

Task 1: Examples of Population and Sample

Below are three everyday examples of **population** and **sample** using apps we use daily:

1. Instagram Followers

Population: All the followers of a particular Instagram account (or all Instagram users in a city/country).

Sample: A randomly selected group of 200 followers whose engagement (likes, comments, story views) is analyzed to understand overall audience behavior.

2. Zomato Restaurants

Population: All restaurants listed on Zomato in a city (e.g., all restaurants in Ahmedabad).

Sample: A selected set of 50 restaurants from different areas of the city whose average delivery time and ratings are studied to estimate overall service quality.

3. Flipkart Products

Population: All products listed under the 'Electronics' category on Flipkart.

Sample: 100 randomly chosen electronic products whose prices, ratings, and return rates are examined to understand customer satisfaction trends.

Task 3: Why Use Sampling for Swiggy Delivery Times?

Collecting delivery time data from every single order ever made on Swiggy would be extremely time-consuming, expensive, and computationally heavy because the total number of orders runs into millions. Sampling allows us to study a representative subset of orders and still draw reliable conclusions about overall delivery performance. It saves resources while providing sufficiently accurate insights for decision-making and improving service quality.

Task 4: Type of Bias in the Spotify Survey Scenario

This scenario introduces **Selection Bias** (specifically a form of **Sampling Bias**).

The survey is sent only to users who recently streamed Bollywood songs. Therefore, the sample is not representative of all Spotify users. People who listen mainly to other genres (Western, Punjabi, Tamil, classical, etc.) are completely excluded. As a result, the findings about music preferences will be skewed toward Bollywood listeners and cannot be generalized to the entire Spotify user base.

Task 5: Two Types of Sampling for BookMyShow Users

To study movie booking habits of BookMyShow users, two suitable sampling methods are:

1. Stratified Sampling

First, divide all BookMyShow users into meaningful strata (groups) based on a relevant characteristic such as city (Mumbai, Delhi, Bangalore, etc.), age group, or preferred language of movies. Then randomly select a proportional number of users from each stratum. This ensures that

users from every major city or age group are properly represented in the sample, giving a more accurate picture of booking habits across different segments.

2. Systematic Sampling

Arrange all registered BookMyShow users in a list (for example, ordered by user ID or registration date). Choose a random starting point and then select every k-th user (where k = total users ÷ desired sample size). For instance, if there are 1,00,000 users and we need a sample of 1,000, we select every 100th user. This method is simple to implement and spreads the sample evenly across the entire user base.